

Joy J. Bayan

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Research Fields: Macroeconomics; Monetary Economics; International Economics; Macroeconometrics

Education

Ph.D. in Economics, Indiana University (expected May 2027) 2021-2027

Graduate Studies in Mathematical Economics, Panthéon-Sorbonne University, Paris, France 2019-2020

B.A. in Global Studies/**M.A.** in Public Policy, University of Minnesota 2012-2017

Research

Job Market Paper: Is the Mean Enough? The Distribution of Inflation Expectations and the Aggregate Phillips Curve

Abstract:

The representative-agent New Keynesian Phillips Curve summarizes inflation expectations by their mean, reducing the cross-sectional distribution of beliefs to a single point forecast. This paper examines whether the mean is a sufficient statistic for inflation dynamics. Using individual responses from the University of Michigan Survey of Consumers from 1990 to 2025, I compare the mean expectation with distributional features summarized by dispersion, directional shifts, and tail risk. Accounting for omitted distributional features—particularly dispersion and tail risk in the full sample—produces a steeper aggregate Phillips Curve while the mean expectation becomes insignificant. During COVID, changes in the distribution begin in the tails before developing into a rightward shift, whereas during the Great Recession a leftward shift occurs before the lower tail expands. Hence, the distribution contains information beyond its mean.

Presentations: Federal Reserve Bank of Philadelphia Research Seminar (Aug 2026), Federal Reserve Bank of Cleveland RISE Conference (Sept 2026), Illinois Economic Association (Oct 2026), Southern Economic Association (Nov 2026), Midwest Macroeconomics Conference (Nov 2026)

Conference Service: American Economic Association, “Inflation and Expectations” Session Chair(2027)

Working Papers

When the Poor Pay More for Groceries: Online Retail Transformation and Monetary Policy with Xudong An (Federal Reserve Bank of Philadelphia)

and Rodney Ramcharan (University of Southern California)

Abstract: Using 100 million grocery retail transactions from 228,519 U.S. households between 2019 and 2023, we show that low-income households experienced 1.4% more grocery price inflation than high-income households in the post-COVID period. After 2022, fresh-produce prices rose 5.3% more for low-income households. The increased adoption of online shopping eliminated geographic pricing advantages, reducing relative savings for low-income households by 4.4%, while the expansion of warehouse clubs increased their price premiums by 6.4%, with monetary policy tightening adding a further 0.11%.

Presentations: Federal Reserve Bank of Philadelphia (2026), Insurance Risk Economic Series(2027)

From Regional to Aggregate: Trade Openness and the Phillips Curve Break After 2000

Abstract: This paper examines whether trade openness can explain the break in the U.S. Phillips curve after 2000. Within an open-economy New Keynesian framework, I use U.S. regional panel data and exploit China’s accession to the World Trade Organization in 2001 as a source of trade-induced structural change. I construct a shift-share instrument using pre-2000 regional employment exposure to tradable industries, and find that greater trade exposure contributed to the flattening of the U.S. Phillips curve

Presentations: Midwest Economic Association (2026), American Economic Association*(2027)

Research Experience

Federal Reserve Bank of Philadelphia 06/2026–present

Management Trainee, Ph.D. Research Intern, Retail Supervisory Modeling Team, SRC

Federal Reserve Bank of San Francisco Summer 2024

Ph.D. Research Intern, Supervision, Risk and Credit Department

- Designed a PCA-based risk index for millions of commercial real estate (CRE) loan records

Research Assistantships, Indiana University 2022–2024
Research Assistant to Professors Yoosoon Chang, Todd Walker, and Mostafa Beshkar
• Conducted research in functional data analysis, proxy vector autoregressions, macroeconomic modeling, and international investment disputes using MATLAB, Julia, and R.

Teaching Experience

Instructor: Fundamentals of Macroeconomics for Business (60 students), Indiana University Fall 2024
Teaching Assistant: Introductory Statistics (Honors), Psychology Statistics, Gender Economics, Economics of Race, Fundamentals of Macroeconomics for Business, Indiana University 2021–2025

Academic Service Experience

Accessibility Coordinator, Economics Department, Indiana University 2025–2026
• Coordinated accessibility accommodations for students and advised faculty on accessibility
Development Coordinator, Graduate Student Government, Indiana University 2023–2024
• Organized professional development workshops and social events for 10,000+ graduate students

Professional Experience

Minnesota Department of Health 2020–2021
Senior Research Analyst, COVID-19 Surveillance
• Developed metrics for Governor’s epidemiological daily briefings for statewide public health response
Minnesota Department of Revenue 2017–2019
Star of the North Fellow, Management Analyst 3, Income and Property Tax Divisions
• Developed automated reporting tools for legislative and fiscal inquiries from 3 million + records
Minnesota State Demographic Center 2017–2018
Star of the North Fellow, Research Analyst, Census 2020 Preparation
• Published 2020 population forecast adopted by Minnesota Legislature for congressional redistricting

Fellowships & Awards

IUB Jubilee Graduate Teaching Award (forthcoming, Fall 2026)
Experiential Learning Scholarship, IU Walter Center (Summer 2026, Fall 2026)
IU Daniel J. Duesterberg Travel Award (2025, 2026)
IU Economics Enrichment Travel Fund (2025, 2026)
IU Global Citizens Exchange Scholarship, O’Neill School (2024)
Summer Award, Center for Applied Economics and Policy Research (2023)
Star of the North Fellowship, State of Minnesota (2017–2018)
Lloyd Short Award for Best Master’s Thesis, University of Minnesota (2017)

Professional Development

2026: Federal Reserve Bank of Cleveland, RISE workshop: Regime-Switching and Nonlinear Modeling
2026: Erdos Institute, Data Science Bootcamp
2025: Economic Machine Learning Lab Quant Conference, Indiana University
2025: Workshop on Estimation of Structural Models, University of Notre Dame
2024: Theory and Experiments in Monetary Economics, Indiana University
2024: Heterogeneous Agent Asset Pricing Workshop, Cleveland State University
2023: Advances in Econometrics, Indiana University

Technical Skills

Programming: R, Python, Stata, MATLAB, Julia, SQL, Dynare, Fortran
Methods: Non-parametric Estimation, Time Series, Panel Data, Structural VAR, Machine Learning
Data tools: Git/GitHub, L^AT_EX, Tableau, R Shiny, SQL Server, Census/BLS/FRED APIs
Languages: English, Russian, Mongolian; French (intermediate), German (intermediate)

References

Gerhard Glomm

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Rodney Ramcharan

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